

claude-windows / 068560d3-1ca2-4387-981a-d2d6bca55414

Metadata

- Source: `claude-windows`
- Kind: `claude`
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\068560d3-1ca2-4387-981a-d2d6bca55414\subagents\workflows\wf\_ef85c152-821\agent-a0bba9ca59454daf8.jsonl`
- SHA-256: `b15434680405c2244d73dfb036da83fa052c05cc86a7457cbccbd4fcdbea4725`
- Source modified: `2026-06-16T06:34:45+00:00`
- Imported at: `2026-07-05T16:48:29+00:00`
- project: `wf\_ef85c152-821`
- session_id: `068560d3-1ca2-4387-981a-d2d6bca55414`

Transcript

1. user (2026-06-16T06:32:58.973Z)

You are a SKEPTICAL reviewer. Confirm or REFUTE this audit finding against the ACTUAL code in the worktree C:\Users\avidu\Projects\badminton-highlight-indexer\.claude\worktrees\gf-doc\ (cite file:line). Decide isMustFix = is this a REAL issue that must be fixed before relying on the suite, vs a non-issue / nice-to-have / already-handled? Default to 'refuted' or 'partial' (isMustFix=false) unless you can confirm it in the code.

FINDING [PRIVACY / de-attribution ? completeness of `\_scan\_forbidden`, surrogate non-reversibility, refusal-gate bypass, raw-trajectory fingerprinting, and honesty of the doc's bounded claims, for the shipped `freeze\_real\_fixture` de-attribution path. / high]: Raw per-frame `(Frame,Visibility,X,Y)` trajectory is committed verbatim for real fixtures ? the de Montjoye spatial fingerprint is fully retained, contradicting doc §4c/TM5
EVIDENCE: freeze_real_fixture writes the FULL per-frame trajectory: golden_fixtures.py:635 `write\_trajectory\_csv(rows, fdir/'trajectory.csv')` where rows come from `\_reset\_trajectory\_rows` (lines 538-549) which passes X/Y through UNTOUCHED ('Visibility/X/Y are passed through untouched'), shifting only the frame index. The committed clean_single_rally/trajectory.csv shows full per-frame X/Y (e.g. `1,1,723.164676,372.202099`). This directly contradicts docs/GOLDEN_REGRESSION_FIXTURES.md §4c 'Raw-trajectory exclusion ? never commit the per-frame (Frame,Visibility,X,Y) CSV' and §4b/TM5 'only ~5 invariant scalars/window'. The time-origin reset severs the absolute-TIMELINE correlation but does nothing to the SPATIAL fingerprint the lens explicitly flags (de Montjoye: 4 points ? 95% re-ID). The whole expected.json minimization-to-5-scalars is undercut because the raw CSV sits right next to it in the same tracked directory.

PROPOSED FIX: Either (a) make the design match the doc: do NOT commit trajectory.csv for cleared_real fixtures ? derive expected.json on the GPU/owner box and commit only the aggregate snapshot, with replay reading from a non-committed/Tier-1 trajectory; or (b) if the raw CSV must be committed for replay, FIX the doc (§4c, TM5, TL;DR 'publish shuttle-only, drop person/pose') to state plainly that the full per-frame shuttle trajectory IS published, re-do the §3/§5 non-attributability analysis on THAT basis (the de Montjoye fingerprint argument applies to the committed coords, not just to a withheld CSV), and confirm the owner accepted THAT risk ? the 2026-06-16 acceptance was made against a doc that claims the raw CSV is excluded.

2. assistant (2026-06-16T06:33:02.344Z)

I'll verify this finding by examining the actual code and documentation in the worktree.

3. user (2026-06-16T06:33:05.632Z)

Your configuration specifies to merge with the ref 'refs/heads/feat/golden-fixtures-doc' from the remote, but no such ref was fetched.

4. user (2026-06-16T06:33:05.707Z)

backend\eval\golden_fixtures.py

5. user (2026-06-16T06:33:05.729Z)

docs\GOLDEN_REGRESSION_FIXTURES.md

6. assistant (2026-06-16T06:33:08.376Z)

Let me read the relevant files.

7. user (2026-06-16T06:33:09.812Z)

```
500
501     def _scan_forbidden(text: str, *, where: str, source_paths: Sequence[str]) -> None:
502         """POSITIVE privacy assertion ($4c, red-team?): raise if ``text`` leaks anything
attributable.
503
504         Scans the written JSON *text* for (a) any ``FORBIDDEN_KEYS`` biometric/identity
substring,
505         (b) a 32-hex MD5 (the ``video_id`` shape), or (c) any identifying token of a source
path
506         (full path / basename / stem) ? excluding the fixed schema filename literals the
writers
507         always emit. The trajectory is shuttle-only by nature; this guards against a
*regression*
508         re-introducing a person/audio/pose stream or an id/path back-channel ? never the
silent default.
509         """
510         low = text.lower()
511         for key in FORBIDDEN_KEYS:
512             if key in low:
513                 raise RefusalError(
514                     f"{where}: forbidden key substring {key!r} found in written output ? "
515                     f"Tier-0 fixtures are shuttle-only; person/pose/gait/face/audio/player
MUST NOT "
516                     f"appear (docs/GOLDEN_REGRESSION_FIXTURES.md $4c).")
517         m = _MD5_RE.search(text)
518         if m:
519             raise RefusalError(
520                 f"{where}: a 32-hex MD5 ({m.group(0)!r}) found in written output ? that is
the "
521                 f"video_id shape and an attribution back-channel; never write it ($4c/$5).")
522         for sp in source_paths:
523             if not sp:
524                 continue
525             p = Path(sp)
526             for token in (sp, p.name, p.stem):
527                 t = (token or "").strip().lower()
528                 # Skip a token that is itself a substring of a schema-literal the writers
always emit
529                 # (e.g. a source stem ``traj`` ? ``trajectory.csv``) ? that collision is not
a leak.
530                 if not t or any(t in lit for lit in _SCHEMA_FILENAME_LITERALS):
531                     continue
532                 if t in low:
533                     raise RefusalError(
534                         f"{where}: source-path token {token!r} found in written output ? the
source "
535                         f"filename/path must never be persisted ($4c metadata strip).")
536
537
538     def _reset_trajectory_rows(points: Sequence[Any], t0_frame: int
539                               ) -> List[Tuple[int, int, float, float]]:
540         """Shift every ``TrajectoryPoint`` frame by ``-t0_frame`` ?
``(Frame,Visibility,X,Y)`` rows.
541
542         Visibility/X/Y are passed through untouched (the shuttle geometry is metric-
```

```

invariant under a
543     pure time shift); only the absolute frame index moves toward 0, severing the match
timeline.
544     """
545     rows: List[Tuple[int, int, float, float]] = []
546     for p in points:
547         rows.append((int(p.frame) - t0_frame, 1 if p.visible else 0, float(p.x),
float(p.y)))
548     rows.sort(key=lambda r: r[0])
549     return rows
550
551
552     def _reset_labels(gts: Sequence[Interval], t0_sec: float) -> List[Interval]:
553         """Shift every label by ``-t0_sec`` (same offset as the trajectory ? metric-
invariant)."""
554         return [(s - t0_sec, e - t0_sec) for s, e in gts]
555
556
557     def freeze_real_fixture(
558         label: str,
559         trajectory_csv_path: Any,
560         labels_csv_path: Any,
561         *,
562         license: str,
563         minors_free: bool,
564         owned: bool,
565         fps: float,
566         frame_width: float,
567         fixtures_dir: Path = FIXTURES_DIR,
568         slug: Optional[str] = None,
569         time_origin_reset: bool = True,
570         source_note: str = "",
571     ) -> Dict[str, Any]:
572         """Turn an OWNED, minors-free real trajectory + golden labels into a committable,
de-attributed
573         Tier-0 fixture ? reusing M1's ``replay_fixture`` for the snapshot. ($4c / $5 / $7
row P1.)
574
575         The refusal gate (M2) makes the unsafe path impossible to reach silently: a fixture
is emitted
576         ONLY when ``minors_free is True`` AND ``owned is True`` AND ``license`` is a non-
blank string.
577         The committed bytes carry NO filename / video_id / match / player / capture-date ?
only an
578         opaque random surrogate slug, the shuttle-only trajectory (time-origin reset by
default), the
579         reset labels, and a ``kind="cleared_real"`` provenance block. A positive privacy
assertion
580         re-reads the written ``expected.json`` + ``provenance.json`` and refuses if anything
attributable
581         slipped in.
582
583         NOTE ($6 honest limits): the random surrogate + reset timeline make the fixture non-
attributable
584         only *relative to the unpublished source* ? it is NOT absolutely anonymous (EDPS v.
SRB). De-
585         attribution does not cure provenance/licensing; this sits BEHIND counsel sign-off
(P2) and the
586         owner-asserted Fork-A / minors / biometric-exclusion flags.
587
588         Returns the recomputed ``expected.json`` snapshot dict (identical to
``replay_fixture``).
589         """
590         # --- (a) REFUSAL GATE (M2) ? the code-enforced provenance hard-fail ($4c / $7 ?)
-----

```

```

591     reasons: List[str] = []
592     if minors_free is not True:
593         reasons.append("minors_free must be True (DPDP child = under 18; behavioural
monitoring "
594                         "of a minor is prohibited ? §3, guardrail 'exclude minors')")
595     if owned is not True:
596         reasons.append("owned must be True (only owner-recorded Fork-A source is safe to
commit; "
597                         "broadcast/scraped = Fork B, do NOT commit ? §3 D1)")
598     if not (isinstance(license, str) and license.strip()):
599         reasons.append("license must be a non-blank string (per-record provenance tag ?
§4c "
600                         "'provenance hard-fail gate')")
601     if reasons:
602         raise RefusalError(
603             "REFUSED to freeze a real Tier-0 fixture ? unsafe provenance (de-attribution
does NOT "
604             "cure provenance/licensing; this gate sits behind counsel sign-off,
§6/P2):\n - "
605             + "\n - ".join(reasons))
606
607     traj_path = Path(str(trajjectory_csv_path))
608     labels_path = Path(str(labels_csv_path))
609
610     # --- (b) opaque RANDOM surrogate slug ? never the video_id MD5, never the source
name ---
611     out_slug = slug if slug is not None else _new_surrogate_slug()
612
613     # --- (c) read via the SHIPPED readers; labels via the STRICT loader (never lenient)
----
614     points = TrackNetRunner.parse_trajectory_csv(str(traj_path))
615     if not points:
616         raise ValueError(f"{traj_path}: no parseable trajectory rows
(Frame,Visibility,X,Y)")
617     gts = load_gt_intervals(str(labels_path), strict=True)
618
619     # --- (d) consistent time-origin reset (default ON; MUST be metric-invariant, §4c)
-----
620     # Subtract the SAME offset from BOTH the trajectory frames and the label seconds:
because
621     # metrics.score is built from pure differences (interval_iou, |start-start|, |end-
end|), the
622     # score is bit-identical with vs without reset ? only the absolute timeline is
severed. The
623     # quality-floor / characterization snapshot is therefore unchanged (asserted in the
tests).
624     if time_origin_reset:
625         t0_frame = min(int(p.frame) for p in points)
626         t0_sec = t0_frame / float(fps)
627     else:
628         t0_frame, t0_sec = 0, 0.0
629     rows = _reset_trajectory_rows(points, t0_frame)
630     reset_gts = _reset_labels(gts, t0_sec)
631
632     # --- (e) write the de-attributed fixture (REUSE M1 writers + replay_fixture)
-----
633     fdir = fixtures_dir / out_slug
634     fdir.mkdir(parents=True, exist_ok=True)
635     write_trajectory_csv(rows, fdir / "trajectory.csv")
636     write_labels_csv(reset_gts, fdir / "labels.csv")
637
638     iou = 0.5
639     prov = {
640         "slug": out_slug,
641         "schema_version": CURRENT_SCHEMA,

```

```

642         "kind": "cleared_real",
643         "label": str(label),
644         "license": license,
645         "minors_free": True,
646         "owned": True,
647         "fps": _round(fps),
648         "frame_width": _round(frame_width),
649         "iou": _round(iou),
650         "time_origin_reset": bool(time_origin_reset),
651         "windowing": {
652             "vt": PINNED_VT,
653             "mg": PINNED_MERGE_GAP,
654             "mwd": PINNED_MIN_WINDOW,
655             "max_win": PINNED_MAX_WINDOW,
656         },
657         "paths": {
658             "trajectory": "trajectory.csv",
659             "labels": "labels.csv",

```

8. user (2026-06-16T06:33:10.954Z)

```

1      # Golden Regression Fixtures ? video-free, non-attributable regression suite
2
3      > **Status:** `IN PROGRESS` ? owner-gated . **Owner:** Avi . **Created:** 2026-06-16 .
**Last updated:** 2026-06-16 . **Legal:** owner **accepted the researched risk** 2026-06-16
(condition §3 stand; this is the owner's informed acceptance, **not** a retained-counsel
opinion)
4      > **Lifecycle:** `DRAFT ? IN PROGRESS ? DONE ? archived` (move to `docs/archives/` when
DONE, per archive policy)
5      > **Tracking anchors:** §7 progress tracker is the source of truth; indexed in
`docs/README.md`; pointer in memory `current-state` (and `docs/NEXT_STEPS.md` once work starts).
6      > **Relation to existing docs:** EXTENDS (does not supersede)
`docs/GOLDEN_DATA_SHARING.md` (#175); reuses the eval stack documented in `docs/EVALUATION.md` /
`docs/EXPERIMENT_HARNESS.md`; quality floor ties to `docs/QUALITY_ITERATIONS.md`.
7      > **Honesty note:** §3 (Legal) is **researched general information, NOT legal advice** ?
it gates on retained counsel (§8). §2/§4/§7 mark which claims are **[verified]** against code vs
**[design]** proposals.
8
9      ---
10
11     ## 0. TL;DR
12
13     Let a contributor on **another machine with no video, GPU, WSL, or DB** run the rally-
segmentation regression suite from **committed structured files**, so refactors and non-quality
PRs get a red/green side-effect signal. Two layers, built once and grown by accretion:
14
15     1. **Mechanics** ? freeze the deterministic pipeline at the **post-detector trajectory
seam** and replay it through the **already-shipped** eval stack (`experiment.py` / `metrics.py` /
`windowing.py` / `regression.py`), with two auto-discovered gates: a
**characterization/snapshot** gate (catches any behaviour change) and a **quality-floor** gate
(catches quality drops).
16     2. **Privacy/legal filter (SEALED-FIXTURES)** ? only **de-attributed, minimized, owner-
source, biometric-free** numbers are ever committed; richer/sensitive streams stay key-gated or
out-of-repo. Non-attributability comes from **deletion-at-source**, not encryption.
17
18     **The convergence that makes this clean:** the **privacy** answer (publish shuttle-only,
drop person/pose streams) and the **determinism** answer (freeze only the pure-stdlib `CONTROL`
path, keep the numpy/BLAS learned scorer off the cross-machine gate) point at the **same minimal
public tier**. That coincidence is the spine of the design.
19
20     **Two findings already actionable, independent of the rest:**
21     - ? **Live de-attribution bug** **[verified]: real player/venue names are committed
today in `eval_baselines/heuristic_lovo_n6.json`,
`eval_baselines/gemini_wrapper_2026-06-10.json`, and `backend/eval/eval_partial_labels.py` (the
last also has a personal `C:/Users/avidu/...` path) ? and in git history. ? **Deliverable P0.**

```

```

22     - ? **Determinism trap** *(verified, corroborated by two independent red-teams)*: the
learned variants (`shuttle_only`, `fusion`) train a numpy/OpenBLAS `DYNAMIC_ARCH` logistic
regression whose decisions flip near the 0.5 threshold across CPUs ? would false-RED on a
contributor's machine. ? the cross-machine gate must be **CONTROL/static-only**.
23
24     ---
25
26     ## 1. Problem & goal
27
28     The rally-quality measurement is split across two machines by capability
(`docs/GOLDEN_DATA_SHARING.md`):
29     - `backend/eval/fusion_golden.py` **extracts** `_golden_features.json` ? needs the
**video + trajectory CSV + labels + GPU/WSL** (GPU box only).
30     - `backend/eval/fusion_compare.py` / `backend/eval/ablation.py` **re-score** those JSONs
on **pure CPU, no video** ? any machine.
31
32     The CPU re-scoring is *already* video-free, but the inputs live in gitignored `output/`
/ private OneDrive, so the regression tests that depend on them **skip everywhere**
(`tests/test_ablation.py::test_litmus_reproduces_gate_a`, `tests/test_fusion_compare.py`). The
committed `eval_baselines/` baselines exist but `regression.py` still scores DB-stored
predictions (needs a pipeline run on the video).
33
34     **Goal:** commit a tiny, version-tagged, **numeric-only, non-attributable** per-fixture
corpus + auto-discovered gates so a clean `git clone` / CI gets a real red/green regression
signal with **zero video/GPU/DB**, while never leaking attributable or sensitive data.
35
36     ---
37
38     ## 2. Decisions locked
39
40     | # | Decision | Source | Implication |
41     |---|-----|-----|-----|
42     | D1 | **Corpus is owner-recorded / owned** | owner answer 2026-06-16 | Legal analysis
takes **Fork A** (clean): no third-party copyright/ToS infringement; concern is trade-
secret/moat + privacy. Strongest non-attributability (no public reference to correlate against).
|
43     | D2 | **Repo private now, may open-source later** | owner answer 2026-06-16 | Design
for **public-grade hygiene from day one** ? git history is permanent; nothing scrubable-later. |
44     | D3 | **Adults, consent not formalized** | owner answer 2026-06-16 | **Person/pose/gait
streams stay OUT of the public tier** (DPDP/GDPR derived-personal-data risk without consent).
Public tier = **shuttle + abstract intervals + non-biometric features only**. |
45     | D4 | **Freeze seam = post-detector trajectory** *(design)* | workflow-1 synthesis |
Widest deterministic surface; reuse the shipped eval stack verbatim; the raw per-frame CSV is
**never committed** (only ~5 aggregate, fps/res-invariant shuttle scalars/window). |
46     | D5 | **Cross-machine gate is CONTROL/static-only** *(design)* | both red-teams
[verified cause] | Learned variants (numpy/BLAS) are **owner-box/Tier-1 only**, asserted with
tolerance, never as a cross-machine byte-snapshot. |
47     | D6 | **Non-attributability = deletion-at-source, encryption = public-only barrier**
*(design)* | workflow-2 synthesis | A contributor who runs the tests necessarily reads
plaintext+key (DRM/analog-hole). So we **delete** attribution, not hide it. Encryption only
stops the no-key public. |
48     | D7 | **Two tiers** *(design)* | workflow-2 synthesis | **Tier-0** public/no-key
(shuttle-only, de-attributed). **Tier-1** key-gated/out-of-repo (person/audio, full corpus),
**skips cleanly when absent**. |
49
50     ---
51
52     ## 3. Legal foundation (researched ? gates on counsel, §8)
53
54     **Core verdict (settled across IN/US/EU):** the derived numbers ? per-frame shuttle
(x,y), rally start/end intervals, audio/court/aggregate-person feature values ? **do NOT inherit
the source video's copyright.** They are uncopyrightable *facts/measurements*, not authored
expression, and not a §101 "derivative work."
55
56     | Jurisdiction | Copyright of the data | Database right | Contract / ToS | Net |

```

```

57 |---|---|---|---|---|
58 | **India** (primary) | Not copyrightable as facts (*EBC v. D.B. Modak*); but
compilations get thin protection on a low "skill/labour" bar (*Burlington*, *OLX v. Padawan*) ?
only bites if **lifted from a 3rd-party dataset**, not self-derived | **None** (no sui-generis
right) ? a real permissiveness edge | Anti-scraping ToS (untested in court) + **IT Act
s.43/s.66** civil/criminal exposure for 3rd-party ingestion | **Owned-source + minors-excluded +
no person/pose = clean** |
59 | **US** | Not copyrightable (*Feist*; §102(b); *NBA v. Motorola*) | None (*Feist*
killed sweat-of-the-brow) | **Dominant risk.** ToS enforceable where copyright/CFAA fail
(*ProCD*, *hiQ* ? injunction + **forced deletion**). Remedy for redistribution is takedown, not
nominal damages | Publishable as facts **iff** owned/permissioned source + no barring ToS + no
person/biometric/minor stream |
60 | **EU / UK** | Not copyrightable (*Football Dataco v. Yahoo*) | **Genuinely
contestable.** *Football Dataco v. Sportradar*: recording pre-existing on-court facts =
**obtaining**, not "creating" ? our "we created it" premise is **overconfident**; harm test =
*CV-Online Latvia* | Ryanair v. PR Aviation: ToS can ban extraction/redistribution even with no
IP right | Get EU+UK advice before redistributing numeric streams at scale |
61
62 **Provenance is the decisive fork.** **Fork A (owner-recorded, unpublished)** ? the only
path safe to commit publicly; collapses to a trade-secret *choice* (Indian breach-of-confidence
/ Contract Act 1872 / IT Act s.72 ? **not** US DTSA) + privacy. **Fork B
(broadcast/YouTube/scraped)** ? do **not** commit; the extraction act needs a license or
now-**contested** US fair use (*Thomson Reuters v. Ross* 2025; USCO May-2025), and unlawful
acquisition independently sinks it (*Bartz v. Anthropic*, ~$1.5B settlement). Caveat: owner
footage **shot at a third party's event** (tournament/club) carries
ticket/venue/organisier/broadcaster terms ? treat as Fork B until cleared.
63
64 **Publish-by-stream (privacy):**
65 - ? **Publish-OK:** shuttle (x,y)+Visibility (inanimate object), abstract rally/GT
**intervals** (identity-stripped), non-biometric audio/court/aggregate-person features detached
from identity.
66 - ?? **Conditional (effectively never in v1):** person bounding boxes that *persistently
single out a player* ? flips to GDPR Art.9 the moment it identifies. This architecture is prone
to that flip ? treat per-player tracking as a tripwire.
67 - ? **Never publish:** pose/skeleton/**gait** (re-identifies at 80?95% even with faces
blurred ? *Weiss et al. 2025*), face crops, any named-player?movement mapping. Matches the
existing "gait/biometrics strictly future (GDPR Art.9)" guardrail.
68 - ? **Minors flip everything:** DPDP "child" = under 18; s.9(3) **prohibits behavioural
monitoring** (the Rule-12 exemptions don't cover this project). A persistent per-player
signature on a minor is, conservatively, prohibited. **Exclude minors** from any persisted
stream beyond inanimate shuttle/interval data.
69
70 **Non-attributability is relative, not absolute.** A per-video trajectory is a
fingerprint (de Montjoye: 4 points ? 95% re-ID). It's only non-attributable because the **Fork-A
source stays unpublished** (no public reference to correlate against). Per *EDPS v. SRB* (CJEU
C-413/23 P, 2025) it **remains personal data to the holder** who can re-run the detector. So:
conditional, revocable (publish the source and the link snaps back), forward-looking (re-assess
as the public-video environment grows). Never market it as "anonymous."
71
72 *Key authorities: *Feist* 499 U.S. 340; *NBA v. Motorola* 105 F.3d 841; 17 U.S.C.
§101/§102(b); *Sega v. Accolade* 977 F.2d 1510; *Football Dataco v. Yahoo* (C-604/10) & v.
Sportradar (C-173/11); *BHB v. William Hill* (C-203/02); *CV-Online Latvia* (C-762/19); *Ryanair v.
PR Aviation* (C-30/14); *Thomson Reuters v. Ross* (D. Del. 2025); *EBC v. D.B. Modak*; DPDP Act 2023
s.3/s.9 + Rules 2025; *EDPS v. SRB* (C-413/23 P). Full URLs in the research artifact (§10).
73
74 ---
75
76 ## 4. Architecture (design)
77
78 ### 4a. Freeze seam + reuse map
79 Freeze at the **trajectory CSV ? pure-Python replay**. One shared helper
`backend/eval/golden_fixtures.py::replay_fixture()` is the *single* code path used by both the
freeze tool and both gates:
80 `parse_trajectory_csv ? windowing.build_windows(preset) ? distill_local.build_candidates
? rally_gate net_crossings ? experiment.predict(CONTROL) ? merge_overlaps ? metrics.score`.

```

```

81     Reuses verbatim: `metrics.score/interval_iou`, `segmentation_metrics`,
`experiment.predict/compare/Variant.spec_hash`, `windowing.WindowingPreset` (default SERVED),
`regression.gt_hash/save_baseline + incommensurability guards`, `eval_stats` (seed=0),
`eval_baselines/` committed-baseline precedent.
82
83     ### 4b. Tiered model (SEALED-FIXTURES)
84     | Tier | Where | Key? | Contents | Test behaviour |
85     |---|---|---|---|---|
86     | Tier-0 | `eval_baselines/fixtures/` (tracked) | none | opaque `fixture_id`,
fps/frame_width (quantized), relative `start`/`end`, 5 shuttle scalars, label, relative
`gts`. No person/audio/pose. | Unconditional ? runs on every PR/fork; turns today's
skipped litmus into a real public gate |
87     | Tier-1 | OneDrive (default) or SOPS-encrypted in-repo | token/age key | full
corpus with person/audio; authoritative ?F1/CI/sign-test + exact Gate-A litmus | Skips
cleanly when key/data absent (generalize `_golden_present()` ? `_tier1_present()` ) |
88
89     A fresh no-key clone is always green; coverage never depends on a secret.
90
91     ### 4c. Anti-attribution measures (Tier-0)
92     - Opaque RANDOM surrogate id (`os.urandom` + a private surrogate?source map kept
off-git) ? not the MD5 `video_id`, and not HMAC(salt, video_id?name) (red-team: brute-
forceable over a tiny known roster if the salt leaks). Strip the `name` field (today = source
filename).
93     - Metadata strip ? no filename, video_id, match/event/player identity, capture date.
94     - Time-origin reset ? subtract per-fixture `t0` from all start/end/gts. Provably
metric-invariant (`metrics.interval_iou` and start/end-error are pure differences) ? bit-
identical scores, severed absolute timeline.
95     - Stream minimization ? Tier-0 keeps shuttle block only; person/audio excluded;
pose/gait/face have no field in the schema by design.
96     - Raw-trajectory exclusion ? never commit the per-frame `(Frame,Visibility,X,Y)`
CSV.
97     - Quantize fps/frame_width ? red-team: at n=6 the raw values are a camera/venue
quasi-identifier.
98     - Provenance hard-fail gate ? the de-attribution pass refuses to emit a public
fixture for any record not tagged owner-recorded(Fork-A) + minors-excluded + biometric-excluded.
99
100    ### 4d. The two gates
101    - Characterization/snapshot ? replay fixture, parse-compare JSON (never byte-
compare ? CRLF/float-format safe; `core.autocrlf=true` here), exact on int/structural fields,
`approx(abs=1e-6)` on floats; stamp + check windowing-preset/label/gt fingerprints
(incommensurable ? loud fail). Catches any behaviour change even at constant F1. CONTROL path
only across machines.
102    - Quality-floor ? score vs committed labels; assert per-video + mean F1 ? floor in
`eval_baselines/regression_baseline.json` (created here; carries a windowing fingerprint so
a served-default change is flagged incommensurable, not silently compared); emit `eval_stats`
paired-delta-CI + sign test (NUMBERS INFORM); add a metrics-vs-base-branch paired delta.
Synthetic-tier floors measure correctness, not field quality (documented).
103
104    ### 4e. Serialization & optional protobuf
105    JSON with `sort_keys=True` + pre-quantized floats + LF is sufficient. Protobuf is
optional (01) ? adopt only if cross-OS byte-stable snapshots are wanted (proto3
`rally.fixture.v1`, codegen + `protoc && git diff --exit-code` guard, `*.pb` binary -text).
Protobuf is encoding, not secrecy (trivially `--decode_raw`).
106
107    ### 4f. Encryption & keys (Tier-1 only)
108    Default = out-of-repo + token (existing OneDrive `RALLY_GOLDEN_DIR` seam ? zero
sensitive bytes in git history, sidesteps the legal "redistribution" act). Alternative = SOPS
+ age (per-recipient, revocable, value-level diffs) for Fork-A-clean-but-private data wanting
in-repo versioning. Rejected: git-crypt (whole-repo re-key), git-lfs (storage ? secrecy). CI
secret exposed only to the Tier-1 job on trusted branches, never to fork-PR runners. The
HMAC/surrogate salt (if used) and the surrogate?source map live with the private corpus,
rotated per release. Document explicitly that Tier-1 crypto defends only the no-key public.
109
110    ---
111

```

```

112    ## 5. Threat model (who's protected vs not)
113
114    | Actor | Protected against | NOT protected (by design / residual) |
115    |---|---|---|
116    | **TM1 no-key public** | map fixture?source/match/player; decrypt Tier-1; confirm a
source even holding the candidate file | read Tier-0 numbers + run shuttle-only harness
(intended; useless-in-isolation) |
117    | **TM2 keyed contributor** | recover filename/video_id/match/face/pose/gait ? *never
written* (survives the analog hole) | read every value the suite reads (unpreventable; not
pretended otherwise) |
118    | **TM3 CI runner** | leaked log can't leak attribution it never had | runs the suite
(intended); Tier-1 secret never on fork runners |
119    | **TM4 malicious keyed insider** | correlation needs a public reference; Fork-A source
is unpublished ? "not reasonably likely" | **residual:** conditional + revocable ? publish/leak
the source later and the link can snap back (forward-looking re-test duty) |
120    | **TM5 re-ID adversary** | source unpublished + only ~5 invariant scalars/window +
reset timeline ? no back-match | becomes live if a Fork-A source is ever published |
121    | **TM6 accidental plaintext commit** | required CI leak-scan rejects MD5/known-
name/non-surrogate/plaintext-SOPS | if the required check is disabled, a leak lands |
122
123    ---
124
125    ## 6. Honest limits ? what a green suite does NOT prove
126
127    - Green proves only that the **deterministic post-trajectory transforms** are unchanged
for the **frozen cases**. It does **not** cover: the WASB/TrackNet detector,
`_probe_fps_width`/cv2 frame-seek, the AI/provider handoff, chunking/re-encode, the ffmpeg
stitch, DB persistence/migrations, or the API filter. *A refactor that breaks decoding or the
detector passes these gates green.* The owner-box `regression.py` DB run remains the only full-
chain check.
128    - Characterization = **behaviour-locking, not correctness** ? it freezes current
behaviour *including current bugs*. "Passing" = "behaviour unchanged", never "behaviour
correct". This caveat must ride in the **test pass/fail message**, not only the docs.
129    - **Synthetic-tier floors = plumbing correctness, NOT field quality.** Never cite
synthetic F1 in `QUALITY_ITERATIONS.md` / the paper as rally-detection performance (tag `kind:
synthetic`).
130    - Tier-0 public coverage is **shuttle-only**;; the person/audio fusion ?F1 and exact
Gate-A values need Tier-1 (skipped on no-key clones).
131    - Data is **not "anonymous"*** absolutely (EDPS v. SRB) ? relative, conditional,
revocable, forward-looking.
132    - Encryption gives **no secrecy from a running contributor**;; protobuf gives **no
secrecy at all**.
133    - De-attribution does **not** cure provenance/licensing ? it sits *behind* the per-
record provenance gate + counsel sign-off.
134    - `--refresh-snapshot` can rubber-stamp a real regression; mitigation is **review of the
visible diff** + a separate, deliberate floor-baseline update. Require a recorded reason.
135
136    ---
137
138    ## 7. Deliverables & progress tracker ? **source of truth**
139
140    Legend: ? Todo . ? In progress . ? Done . ? Blocked/gated. One **small PR per row**,
branched from `origin/master`, no stacking. Update the row (status + PR link) as work lands.
141
142    | ID | Deliverable | Depends on | Counsel-gated? | Status | PR |
143    |---|-----|-----|-----|-----|---|
144    | **P0** | **Leak remediation + required CI leak-scan.** Scrub real names/paths in
`eval_baselines/heuristic_lovo_n6.json`, `gemini_wrapper_2026-06-10.json`,
`backend/eval/eval_partial_labels.py`, `tests/test_cleanup_caches.py` ? opaque surrogate keys +
private map; add CI leak-scan (MD5 / known-name / non-surrogate / plaintext-SOPS) as a
**required status check**;; decide on git history (PR #77, accept vs `filter-repo`). | ? | No
(owned data hygiene) | ? | ? |
145    | **M1** | **Fixture framework + synthetic Tier-0 + shared replay helper** (also
delivered M2-core + M3 ? see below). `backend/eval/golden_fixtures.py` (`replay_fixture`, schema
dispatch, `Path(__file__)`-relative) + 3 **tool-generated** synthetic fixtures + manifest +

```

README + `.gitattributes` LF pin + `tests/test\_golden\_fixtures.py`. CONTROL-only / pure-stdlib; cross-OS verified. | ? | No (synthetic) | ? | [#189](https://github.com/avidullu/badminton-highlight-indexer/pull/189) |

146 | **M2** | **Freeze/refresh extractor.** Core (`freeze\_synthetic` / `refresh\_snapshot` / `--check` CLI + idempotence test) shipped in M1 (#189); the `--license`/`--minors-free`/`--owned` **refusal gate** shipped with P1 (this PR). | M1 | No | ? |

[#189](https://github.com/avidullu/badminton-highlight-indexer/pull/189) · **this PR** |

147 | **M3** | **Characterization gate** ? `tests/test\_golden\_fixtures.py`: parse-compared (exact-int / approx-float 1e-6), **CONTROL-only**, positive no-person/audio assertion, perturbation + idempotence. **Shipped in M1 (#189).** (Windowing/label fingerprint guard + row-count cap ? fold into M4.) | M1 | No | ? | [#189](https://github.com/avidullu/badminton-highlight-indexer/pull/189) |

148 | **MX** | **Cross-OS determinism guard (CI).** Lightweight `golden-crossos` CI job (ubuntu/windows/macOS) running `--check` + the fixtures test ? locks the cross-OS guarantee (CONTROL replay is numpy-only, no GPU stack; a *separate fast job*, not full-suite×3). | M1 | No | ? | [#191](https://github.com/avidullu/badminton-highlight-indexer/pull/191) |

149 | **M4** | **Quality-floor gate + first `regression\_baseline.json`.** **tests/test_golden_quality_floor.py**; create `eval\_baselines/regression\_baseline.json` with **windowing fingerprint**; metrics-vs-base paired-delta; corpus-level (not per-slug) LOVO for runtime. | M1, M2 | No | ? | ? |

150 | **P1** | **De-attribution + minimization** ? `golden\_fixtures.freeze\_real\_fixture()` (extends the M1 module, not a separate script): **opaque random `fx\_hex` surrogate slug**, metadata strip, metric-invariant **time-origin reset**, strict GT loader, **positive no-person/audio/MD5/source-path assertion**, manifest non-clobber; **freeze-real** CLI with the **refusal gate** (exit 2 unless owned + minors-free + license). Tooling-only, 12 synthetic tests; **no real data committed**. | M1 | No (tooling) | ? | **this PR** |

151 | **P2** | **Owner-accepted (researched-risk), 2026-06-16.** Owner accepted the researched IP/privacy analysis (§3) to proceed **under its conditions** (owned/Fork-A source · minors excluded · biometric/person streams excluded · de-attributed). Owner's informed risk acceptance ? **NOT a retained-counsel opinion**; optional future counsel confirmation per §8. | P1 | Owner-accepted | ? | ? |

152 | **P3** | **Tier-0 REAL subset.** Commit de-attributed, minors/biometric-excluded, owned Fork-A real fixtures (unblocked by P2). Hard-gated **in code** by P1's refusal gate (owned + minors-free + license required) ? needs the owner's per-video clearance list. | P1, P2 | conditions §3 | ? | ? |

153 | **O1** | **(optional) Protobuf + deterministic serialization.** `rally.fixture.v1` proto3, codegen + `protoc`/`git diff --exit-code` guard, canonical writer, **gitattributes** binary. Only if cross-OS byte-stable snapshots are wanted. | M1 | No | ? | ? |

154 | **O2** | **(optional) Tier-1 key-gated path.** `scripts/fetch\_golden.py` (OneDrive token, default) and/or SOPS+age; **tier1_present()**; **RALLY_GOLDEN_DIR** wired; Tier-1 CI job gated on secret, never on fork runners. | M1 | No | ? | ? |

155 | **DOC** | **Discoverability cross-links** ? added the **eval_baselines/fixtures/** row to **CODE_MAP §0** + a **golden_fixtures.py** entry to **CODE_MAP §2.3**; cross-linked **GOLDEN_DATA_SHARING.md** (committed subset ? private OneDrive corpus). README indexed in #186; memory **current-state** kept live. (EVALUATION/QUALITY_ITERATIONS cross-links **deferred** ? that doc surface is hot in the concurrent quality track; avoid collision.) | M1 | No | ? | **this PR** |

156

157 **Must-fix-before-any-public-commit (red-team), folded into the rows above:** ? numeric-tolerance not byte-equality + multi-OS matrix + CONTROL-only (M3/D5/MX ?); ? scrub names in **all** files + history decision (P0); ? leak-scan a **required** check, hooks are convenience only (P0); ? person-optional loader + **positive** no-person assertion, don't trust the silent `0.0` default (P1 ?); ? random surrogate id not HMAC-over-roster (P1 ?); ? quantize/drop fps/frame_width (P1 ? fps/fw retained as scale constants; quantize is a P3 option); ? document Tier-0 omits the fusion path (DOC); ? counsel + Fork-A + minors/biometric exclusion as hard preconditions (P2 owner-accepted; enforced in code by P1's refusal gate ?).

158

159 **Progress log (quantitative ? coverage must hold/improve, CI incl. the 3-OS cross-OS job green)**

160 | PR | Deliverable(s) | Suite | Coverage | Notes |

161 | ----|-----|-----|-----|-----|

162 | [#189](https://github.com/avidullu/badminton-highlight-indexer/pull/189) | M1 + M2-core + M3 | 746 pass / 7 skip | 80.41% | synthetic Tier-0 + CONTROL replay + characterization; cross-OS proven (CI Linux replayed windows-frozen fixtures) |

163 | [#191](https://github.com/avidullu/badminton-highlight-indexer/pull/191) | MX +

```

tracker/legal reconcile | full suite green | 80.41% | `golden-crossos` CI job green on
ubuntu+windows+macos ? cross-OS **enforced** ; legal ? owner-accepted |
164 | _this PR_ | P1 + M2 refusal gate | 759 pass / 7 skip | **80.49%** (+0.08) |
`freeze_real_fixture` + refusal gate; **13 tests** ; tooling-only, no real data committed |
165
166 ---
167
168 ## 8. Open owner / counsel questions
169
170 **Counsel questions ? documented basis (the owner accepted the researched risk on
2026-06-16; see P2). These remain the record, and a retained-counsel confirmation is
optional/recommended before scaling beyond a small owned, minors-free, de-attributed subset:**
171 1. Does the EU/UK **sui-generis database right** subsist in our self-derived
shuttle/rally streams on the *Sportradar* "obtained vs created" reasoning, and does
redistributing a de-attributed Tier-0 subset cause *CV-Online* harm?
172 2. Post *Thomson Reuters v. Ross* + USCO-2025, is **US fair use for the extraction
step** defensible if the dataset could compete in a "data/analytics" market ? does publishing
only numbers (never footage) matter?
173 3. India: are anti-scraping ToS enforceable; **IT Act s.43/s.66** exposure for any 3rd-
party ingestion; does pending *ANI v. OpenAI* (~Apr 2026) move it? Confirm **DPDP s.9/Rule 10**
and whether per-player tracking is prohibited s.9(3) monitoring of a minor.
174 4. Sign-off on the **final committed Tier-0 subset + provenance log** ; is non-
attributability-via-private-source a defensible posture to assert given its
conditional/revocable nature?
175
176 **Owner (strategic):**
177 5. Publishing the Fork-A subset **forfeits the trade-secret moat** (Indian breach-of-
confidence framing) ? weighed against community/benchmark/paper aims. Your call.
178 6. **Forward-looking commitment:** do not publish a highlight reel of the *same* source
whose fixtures are committed (without re-clearing), and accept a periodic re-assessment as the
public-video environment grows.
179 7. **Scope of first public commit:** synthetic-only (zero provenance risk, recommended)
vs a counsel-cleared real subset.
180 8. **Tier-1 mechanism:** out-of-repo+token (recommended) vs SOPS+age in-repo. And: is
protobuf (01) wanted, or is canonical-JSON enough?
181
182 ---
183
184 ## 9. Definition of done
185
186 The project is **DONE** (flip `Status ? DONE`, archive) when:
187 - [ ] **P0, M1?M4, P1, DOC** are ? and merged.
188 - [ ] A clean `git clone` on **win + ubuntu + mac** runs `python run_all_tests.py` and
the **Tier-0 gates pass green with no key / no video / no GPU / no DB**.
189 - [ ] The **CI leak-scan is a required status check** and passes (no MD5 / real-name /
non-surrogate / plaintext-SOPS in the tree).
190 - [ ] Characterization + quality-floor gates are **parse-compared, CONTROL-only,
windowing-fingerprinted**, and demonstrably catch a seeded behaviour change.
191 - [ ] `docs/CODE_MAP.md §0` + TEST MAP updated; this doc cross-linked; `NEXT_STEPS.md` +
memory `current-state` point here.
192 - [ ] **Either** the public commit is synthetic-only, **or** (**P2 owner-accepted ? +
P3**) for a real-data commit, under the §3 conditions enforced **in code** by P1's refusal gate.
193 - [ ] §6 honest-limits caveat rides in the test pass/fail messages, not only the docs.
194
195 *Optional 01/02 may remain ? without blocking DONE (they're earned, not assumed).*
196
197 ---
198
199 ## 10. References
200
201 **Internal code anchors (verified):** `backend/eval/{regression.py,fusion_golden.py,fusi
on_compare.py,ablation.py,experiment.py,metrics.py>windowing.py,distill_local.py,gt_loader.py,ev
al_stats.py}` · `backend/pipeline/detectors/{base.py,tracknet_runner.py,rally_gate.py}` ·
`backend/utils/hashing.py::compute_video_id` · `eval_baselines/{heuristic_lovo_n6.json,gemini_wr
apper_2026-06-10.json,experiment_leaderboard.json}` ·

```

```

`tests/test_ablation.py::test_litmus_reproduces_gate_a` · `github/workflows/ci.yml`.
202     **Internal docs:** `GOLDEN_DATA_SHARING.md` (#175), `EVALUATION.md`,
`EXPERIMENT_HARNESS.md`, `QUALITY_ITERATIONS.md`, `MULTI_SIGNAL_FUSION_PLAN.md`, `CODE_MAP.md`
$0`.
203     **Legal authorities:** see §3 (full URLs in the research workflow artifact). *This
section is researched general information, not legal advice.*
204
205     ---
206
207     ### Changelog
208     - **2026-06-16** ? Initial draft synthesized from two multi-agent workflows (mechanics
design + 2 critiques; legal memo IN/US/EU + SEALED-FIXTURES architecture + red-team). Verified
the live name-leak and the BLAS-determinism cause against code. Status: DRAFT, owner-gated.
209

```

9. assistant (2026-06-16T06:33:15.130Z)

This is a nuanced finding about a design/doc-vs-code consistency issue. Let me read the rest of `freeze\_real\_fixture` and check whether the committed fixtures actually include real (non-synthetic) raw trajectories, plus examine the doc claims carefully.

10. user (2026-06-16T06:33:17.075Z)

```

659         "labels": "labels.csv",
660         "expected": "expected.json",
661     },
662     "gt_hash": gt_hash(reset_gts),
663     "source_note": str(source_note),
664     "note": (
665         "De-attributed REAL Fork-A fixture (owner-recorded, owned, minors-excluded,
biometric-"
666         "free, shuttle-only). Opaque RANDOM surrogate slug; absolute timeline reset.
NON-"
667         "attributable only relative to the UNPUBLISHED source (EDPS v. SRB) ? never
'anonymous'. "
668         "Tier-0 / CONTROL-only. §6: behaviour-locking, not correctness; sits behind
counsel (P2).",
669     ),
670 }
671 _write_json(fdir / "provenance.json", prov)
672
673 expected = replay_fixture(fdir)          # REUSE M1 ? correct-by-construction
snapshot
674 _write_json(fdir / "expected.json", expected)
675
676 # --- (f) POSITIVE privacy assertion AFTER writing (scan the written text, §4c / ?)
-----
677 source_paths = [str(traj_path), str(labels_path)]
678 _scan_forbidden((fdir / "expected.json").read_text(encoding="utf-8"),
679                 where=f"{out_slug}/expected.json", source_paths=source_paths)
680 _scan_forbidden((fdir / "provenance.json").read_text(encoding="utf-8"),
681                 where=f"{out_slug}/provenance.json", source_paths=source_paths)
682
683 # --- (g) append to the manifest with kind="cleared_real" (don't clobber synthetic)
-----
684 _append_manifest_entry(prov, fixtures_dir)
685 return expected
686
687
688 def _manifest_entry_for(prov: Dict[str, Any]) -> Dict[str, Any]:
689     """A manifest row for a single fixture, derived from its provenance block."""
690     slug = prov["slug"]
691     return {
692         "slug": slug,
693         "fps": _round(float(prov["fps"])),

```

```

694         "frame_width": _round(float(prov["frame_width"])),
695         "windowing": dict(prov.get("windowing", {})),
696         "kind": prov.get("kind", "synthetic"),
697         "schema_version": int(prov.get("schema_version", CURRENT_SCHEMA)),
698         "paths": {
699             "dir": slug,
700             "trajectory": f"{slug}/trajectory.csv",
701             "labels": f"{slug}/labels.csv",
702             "expected": f"{slug}/expected.json",
703             "provenance": f"{slug}/provenance.json",
704         },
705     }
706
707
708     def _append_manifest_entry(prov: Dict[str, Any], fixtures_dir: Path) -> None:
709         """Add/replace one fixture's manifest row WITHOUT clobbering the others (e.g.
710         synthetic)."""
711         entries = load_manifest(fixtures_dir)
712         entries = [e for e in entries if e.get("slug") != prov["slug"]]
713         entries.append(_manifest_entry_for(prov))
714         entries.sort(key=lambda e: str(e.get("slug")))
715         _write_json(fixtures_dir / "manifest.json", entries)
716
717     # =====
718     # Manifest + loaders
719     # =====
720
721
722     def _write_manifest(specs: Sequence[FixtureSpec], fixtures_dir: Path) -> None:
723         entries = [{
724             "slug": spec.slug,
725             "fps": _round(spec.fps),
726             "frame_width": _round(spec.frame_width),
727             "windowing": {
728                 "vt": PINNED_VT, "mg": PINNED_MERGE_GAP,
729                 "mwd": PINNED_MIN_WINDOW, "max_win": PINNED_MAX_WINDOW,
730             },
731             "kind": "synthetic",
732             "schema_version": CURRENT_SCHEMA,
733             "paths": {
734                 "dir": spec.slug,
735                 "trajectory": f"{spec.slug}/trajectory.csv",
736                 "labels": f"{spec.slug}/labels.csv",
737                 "expected": f"{spec.slug}/expected.json",
738                 "provenance": f"{spec.slug}/provenance.json",
739             },
740         } for spec in specs]
741         _write_json(fixtures_dir / "manifest.json", entries)
742
743
744     def load_manifest(fixtures_dir: Path = FIXTURES_DIR) -> List[Dict[str, Any]]:
745         """Load the fixtures manifest (returns [] if absent)."""
746         path = fixtures_dir / "manifest.json"
747         if not path.is_file():
748             return []
749         data = _read_json(path)
750         if not isinstance(data, list): # pragma: no cover - corrupt manifest
751             raise ValueError(f"{path}: manifest must be a JSON array")
752         return data
753
754
755     def load_fixture(slug: str, fixtures_dir: Path = FIXTURES_DIR) -> Dict[str, Any]:
756         """Load one fixture's committed artifacts: ``{provenance, expected, dir}``. Refuses
757         a fixture

```

```

757     whose ``schema_version`` exceeds ``CURRENT_SCHEMA`` (forward-incompatible)."""
758     fdir = fixtures_dir / slug
759     prov = _read_json(fdir / "provenance.json")
760     expected = _read_json(fdir / "expected.json")
761     sv = int(expected.get("schema_version", prov.get("schema_version", 0)))
762     if sv > CURRENT_SCHEMA:
763         raise ValueError(
764             f"{slug}: schema_version {sv} > CURRENT_SCHEMA {CURRENT_SCHEMA} ? upgrade
the harness")
765     return {"slug": slug, "dir": fdir, "provenance": prov, "expected": expected}
766
767
768     # =====
769     # Parse-compare (NEVER byte-compare): exact ints/structure, abs-tol floats
770     # =====
771
772
773     def compare_expected(recomputed: Dict[str, Any], committed: Dict[str, Any]) ->
List[str]:
774         """Return a list of human-readable mismatch messages (empty == match).
775
776         Parse-compare per HARD CONSTRAINT 2: structural/int fields (candidate count,
net_crossings,
777         TP/FP/FN, segment count, num_pred/num_gt, slug, schema_version, gt_hash) must be
EXACTLY
778         equal; float fields compared with abs tolerance 1e-6. Never a byte compare ?
CRLF/float-format

```

11. user (2026-06-16T06:33:17.324Z)

```

eval_baselines\fixtures\README.md
eval_baselines\fixtures\clean_single_rally\expected.json
eval_baselines\fixtures\clean_single_rally\labels.csv
eval_baselines\fixtures\clean_single_rally\provenance.json
eval_baselines\fixtures\clean_single_rally\trajectory.csv
eval_baselines\fixtures\manifest.json
eval_baselines\fixtures\multi_rally_with_deadtime\expected.json
eval_baselines\fixtures\multi_rally_with_deadtime\labels.csv
eval_baselines\fixtures\multi_rally_with_deadtime\provenance.json
eval_baselines\fixtures\multi_rally_with_deadtime\trajectory.csv
eval_baselines\fixtures\occlusion_break\expected.json
eval_baselines\fixtures\occlusion_break\labels.csv
eval_baselines\fixtures\occlusion_break\provenance.json
eval_baselines\fixtures\occlusion_break\trajectory.csv

```

12. assistant (2026-06-16T06:33:23.615Z)

Now this is critical. The finding cites `clean\_single\_rally/trajectory.csv` as evidence that "real" raw trajectory is committed. But the glob shows only THREE fixtures: `clean\_single\_rally`, `multi\_rally\_with\_deadtime`, `occlusion\_break`. The doc (M1, line 145) says these are "3 tool-generated synthetic fixtures." Let me verify the `kind` of these committed fixtures and confirm whether ANY `cleared\_real` fixture exists.

13. user (2026-06-16T06:33:24.571Z)

```

1      [
2      {
3          "fps": 30.0,
4          "frame_width": 1280.0,
5          "kind": "synthetic",
6          "paths": {
7              "dir": "clean_single_rally",
8              "expected": "clean_single_rally/expected.json",
9              "labels": "clean_single_rally/labels.csv",

```

```

10         "provenance": "clean_single_rally/provenance.json",
11         "trajectory": "clean_single_rally/trajectory.csv"
12     },
13     "schema_version": 1,
14     "slug": "clean_single_rally",
15     "windowing": {
16         "max_win": 0.0,
17         "mg": 2.0,
18         "mwd": 2.0,
19         "vt": 0.02
20     }
21 },
22 {
23     "fps": 30.0,
24     "frame_width": 1280.0,
25     "kind": "synthetic",
26     "paths": {
27         "dir": "multi_rally_with_deadtime",
28         "expected": "multi_rally_with_deadtime/expected.json",
29         "labels": "multi_rally_with_deadtime/labels.csv",
30         "provenance": "multi_rally_with_deadtime/provenance.json",
31         "trajectory": "multi_rally_with_deadtime/trajectory.csv"
32     },
33     "schema_version": 1,
34     "slug": "multi_rally_with_deadtime",
35     "windowing": {
36         "max_win": 0.0,
37         "mg": 2.0,
38         "mwd": 2.0,
39         "vt": 0.02
40     }
41 },
42 {
43     "fps": 30.0,
44     "frame_width": 1280.0,
45     "kind": "synthetic",
46     "paths": {
47         "dir": "occlusion_break",
48         "expected": "occlusion_break/expected.json",
49         "labels": "occlusion_break/labels.csv",
50         "provenance": "occlusion_break/provenance.json",
51         "trajectory": "occlusion_break/trajectory.csv"
52     },
53     "schema_version": 1,
54     "slug": "occlusion_break",
55     "windowing": {
56         "max_win": 0.0,
57         "mg": 2.0,
58         "mwd": 2.0,
59         "vt": 0.02
60     }
61 }
62 ]
63

```

14. user (2026-06-16T06:33:25.501Z)

```

1     {
2         "description": "One continuous ~4s rally; shuttle oscillates across the net ? 1
window.",
3         "fps": 30.0,
4         "frame_width": 1280.0,
5         "iou": 0.5,
6         "kind": "synthetic",
7         "license": "synthetic",

```

```

8      "minors_free": true,
9      "note": "Synthetic, deterministic, owned (no real footage). Tier-0 / CONTROL-only.
Plumbing-correctness fixture ? NOT a measure of rally-detection quality (§6).",
10     "paths": {
11         "expected": "expected.json",
12         "labels": "labels.csv",
13         "trajectory": "trajectory.csv"
14     },
15     "schema_version": 1,
16     "slug": "clean_single_rally",
17     "windowing": {
18         "max_win": 0.0,
19         "mg": 2.0,
20         "mwd": 2.0,
21         "vt": 0.02
22     }
23 }
24

```

15. user (2026-06-16T06:33:26.349Z)

```

1      Frame,Visibility,X,Y
2      0,1,640.000000,360.000000
3      1,1,723.164676,372.202099
4      2,1,802.694657,382.294345
5      3,1,875.114101,388.531695
6      4,1,937.257930,389.835657
7      5,1,986.410162,385.980762
8      6,1,1020.422607,377.633558
9      7,1,1037.808758,366.237351
10     8,1,1037.808758,353.762649
11     9,1,1020.422607,342.366442
12     10,1,986.410162,334.019238
13     11,1,937.257930,330.164343
14     12,1,875.114101,331.468305
15     13,1,802.694657,337.705655

```

16. assistant (2026-06-16T06:33:36.447Z)

This is the crux. The finding's central factual claim is wrong. Let me confirm the key points:

1. The fixture `clean\_single\_rally` that the finding cites as evidence (`1,1,723.164676,372.202099`) is ``kind="synthetic", `license="synthetic", "Synthetic, deterministic, owned (no real footage)"` ? NOT a `cleared\_real` fixture. The X/Y coordinates are mathematically generated (frame 0 = 640,360 exactly = center; a smooth parabolic arc), not extracted from a real video.
2. There are ZERO `cleared\_real` fixtures committed. The manifest confirms all three are `synthetic`.
3. The finding claims "Raw per-frame trajectory is committed verbatim for real fixtures" ? but `freeze\_real\_fixture` is tooling only and no real fixtures have been frozen/committed. The doc (line 150, P1) explicitly states "Tooling-only ... no real data committed" and P3 (line 152, the actual real-data commit) is status ? Todo.

Now let me check the doc's actual claims about the raw CSV vs. what the tooling does, to assess the doc-vs-code consistency angle (which is the genuinely interesting part of the finding).

The finding quotes §4c line 96: "Raw-trajectory exclusion ? never commit the per-frame `(Frame,Visibility,X,Y)` CSV." And §4b line 86 / TM5 line 120: Tier-0 = "5 shuttle scalars" / "~5 invariant scalars/window."

Let me verify whether `freeze\_real\_fixture` indeed writes the per-frame CSV (it does, line 635) and whether `expected.json` contains per-frame data or aggregates. Let me check `replay\_fixture` and the expected.json contents.

17. user (2026-06-16T06:33:37.405Z)

```
1      {
2        "candidates": [
3          {
4            "end": 3.966667,
5            "shuttle": {
6              "active_ratio": 1.008475,
7              "duration": 3.933333,
8              "mean_velocity": 1.27678,
9              "net_crossings": 7,
10             "peak_velocity": 1.970041
11           },
12           "start": 0.033333
13         }
14       ],
15       "control_segments": [
16         [
17           0.033333,
18           3.966667
19         ]
20       ],
21       "gt_hash": "6c93801fd607",
22       "iou": 0.5,
23       "schema_version": 1,
24       "score": {
25         "f1": 1.0,
26         "false_negatives": 0,
27         "false_positives": 0,
28         "iou_threshold": 0.5,
29         "mean_end_error": 0.033333,
30         "mean_iou": 0.983333,
31         "mean_start_error": 0.033333,
32         "num_gt": 1,
33         "num_pred": 1,
34         "precision": 1.0,
35         "recall": 1.0,
36         "true_positives": 1
37       },
38       "slug": "clean_single_rally"
39     }
40
```

18. user (2026-06-16T06:33:38.714Z)

```
296: def replay_fixture(slug_or_dir: Any) -> Dict[str, Any]:
297-     """Replay one fixture through the shipped CONTROL path; return its recomputed snapshot.
298-
299-     Resolves ``slug_or_dir`` (a slug string or a fixture directory ``Path``) to its
300-     committed
301-     ``trajectory.csv`` + ``labels.csv`` + ``provenance.json``, then:
302-
303-     parse_trajectory_csv ? build_candidates(proposal=pinned) ? VideoCandidates(5 shuttle
304-     cols only) ? predict(CONTROL, vc) ? score(preds, gts, iou).
305-
306-     Returns ``{slug, schema_version, iou, candidates:[{start,end,shuttle:{5 names}}],
307-     control_segments:[[s,e]], score:{...}, gt_hash}``. CONTROL keeps every candidate then
308-     merges overlaps; NO numpy is touched.
309-
310-     """
311-     fdir = _resolve_dir(slug_or_dir)
312-     slug = fdir.name
313-     traj_csv = fdir / "trajectory.csv"
314-     labels_csv = fdir / "labels.csv"
315-     prov = _read_json(fdir / "provenance.json") if (fdir / "provenance.json").is_file() else
316-     None
```

```

314-
315-     fps, fw = _fps_fw(prov, slug)
316-     iou = float(prov["iou"]) if (prov and "iou" in prov) else 0.5
317-     proposal = _pinned_proposal(prov)
318-
319-     intervals, feature_rows = distill_local.build_candidates(
320-         str(traj_csv), fps=fps, frame_width=fw, proposal=proposal)
321-
322-     # Build VideoCandidates DIRECTLY with ONLY the 5 shuttle feature columns. We never call
323-     # fusion_compare.load_videos / ablation.load_videos ? those read c['person']
324-     # and would KeyError on these person-free fixtures (HARD CONSTRAINT 1).
325-     features: Dict[str, List[float]] = {
326-         name: [row[i] for row in feature_rows] for i, name in enumerate(SHUTTLE_FEATURES)
327-     }
328-     gts = load_gt_intervals(str(labels_csv), strict=True)
329-     vc = VideoCandidates(name=slug, intervals=intervals, features=features, gts=gts)
330-
331-     preds = predict(CONTROL, vc) # CONTROL: keep all ? merge_overlaps; pure-stdlib, no
332-     # numpy
333-     sc = score(preds, gts, iou)
334-
335-     candidates: List[Dict[str, Any]] = []
336-     for j, (s, e) in enumerate(intervals):
337-         shuttle = {name: _round(feature_rows[j][i]) for i, name in
338-             enumerate(SHUTTLE_FEATURES)}
339-         # net_crossings is conceptually an integer count ? keep it exact (int), per
340-         # constraint 2.
341-         shuttle["net_crossings"] =
342-             int(round(feature_rows[j][SHUTTLE_FEATURES.index("net_crossings")]))
343-         candidates.append({"start": _round(s), "end": _round(e), "shuttle": shuttle})
344-
345-     return {
346-         "slug": slug,
347-         "schema_version": CURRENT_SCHEMA,
348-         "iou": _round(iou),
349-         "candidates": candidates,
350-         "control_segments": [[_round(s), _round(e)] for s, e in preds],
351-         "score": {
352-             "iou_threshold": _round(sc.iou_threshold),
353-             "num_pred": int(sc.num_pred),
354-             "num_gt": int(sc.num_gt),
355-             "true_positives": int(sc.true_positives),
356-             "false_positives": int(sc.false_positives),
357-             "false_negatives": int(sc.false_negatives),
358-             "precision": _round(sc.precision),
359-             "recall": _round(sc.recall),
360-             "f1": _round(sc.f1),
361-             "mean_iou": _round(sc.mean_iou),
362-             "mean_start_error": _round(sc.mean_start_error),
363-             "mean_end_error": _round(sc.mean_end_error),
364-         },
365-         "gt_hash": gt_hash(gts),
366-     }
367-
368-# =====
369-# Freeze / refresh (expected.json + provenance.json are correct-by-construction)

```

19. assistant (2026-06-16T06:33:50.399Z)

Now I have the full picture. This confirms the architecture: `expected.json` contains the ~5 aggregated shuttle scalars per window (active_ratio, duration, mean_velocity, net_crossings, peak_velocity), matching the doc's "5 shuttle scalars" claim. But the raw per-frame `trajectory.csv` (the `(Frame,Visibility,X,Y)` CSV) ALSO sits in the same tracked directory and

IS committed ? for the synthetic fixtures, and `freeze\_real\_fixture` would write it for real ones too (line 635).

So the finding has identified a REAL doc-vs-code inconsistency, but mis-stated the headline. Let me assess the core technical claim about whether the doc says the raw CSV is "never committed."

The doc's claims:

- §4c line 96: "***Raw-trajectory exclusion** ? never commit the per-frame `(Frame,Visibility,X,Y)` CSV."
- §4b line 86 (Tier-0 contents): "***5 shuttle scalars**... No person/audio/pose."
- §2 D4 line 45: "the raw per-frame CSV is ***never committed** (only ~5 aggregate, fps/res-invariant shuttle scalars/window)."
- TM5 line 120: "only ~5 invariant scalars/window + reset timeline ? no back-match"

The actual shipped code (`replay\_fixture`, `freeze\_synthetic`, `freeze\_real\_fixture`) ALL write `trajectory.csv` containing the full per-frame `(Frame,Visibility,X,Y)` rows into the tracked `eval\_baselines/fixtures/<slug>/` directory. The synthetic fixtures already demonstrate this ? `clean\_single\_rally/trajectory.csv` is committed with full per-frame coords. So the doc's "raw-trajectory exclusion ? never commit the per-frame CSV" statement is contradicted by the actual M1-shipped design where the replay reads the committed `trajectory.csv`.

Let me verify the README in the fixtures dir to see if it acknowledges the committed CSV, and double-check the doc doesn't reconcile this elsewhere.

20. user (2026-06-16T06:33:51.827Z)

```
1      # `eval_baselines/fixtures/` ? Golden Regression Fixtures (Tier-0, synthetic)
2
3      Video-free, **CONTROL-only** characterization fixtures for the rally-segmentation
pipeline.
4      A contributor on **any machine ? no video, GPU, WSL, or DB** ? replays these committed
5      structured files through the already-shipped eval stack and gets a red/green regression
6      signal. See `docs/GOLDEN_REGRESSION_FIXTURES.md` (deliverable **M1**; §4, §6, §7).
7
8      ## What's here
9
10     Each fixture is one directory:
11
12     | File | Purpose |
13     |-----|-----|
14     | `trajectory.csv` | Synthetic shuttle trajectory `(Frame,Visibility,X,Y)` ? the freeze
seam input. |
15     | `labels.csv` | Synthetic ground-truth rally intervals `(start,end)`. |
16     | `expected.json` | Frozen replay snapshot (candidates + 5 shuttle features + CONTROL
segments + score). |
17     | `provenance.json` | Pinned windowing params + privacy/licensing tags + description. |
18
19     `manifest.json` lists every fixture; `backend/eval/golden_fixtures.py` is the shared
replay
20     module + freeze/refresh/check CLI.
21
22     ## Guardrails (non-negotiable)
23
24     - **Synthetic / owned only.** Every fixture is generated by deterministic formulae (NO
random)
25     ? zero third-party footage, zero provenance/licensing risk. Tagged `kind:
"synthetic"`,
26     `license: "synthetic"`, `minors_free: true`.
27     - **No biometric / identity streams.** Tier-0 carries the **5 shuttle feature columns
only**
28     (`duration, peak_velocity, mean_velocity, active_ratio, net_crossings`). There is, by
design,
29     **no person / pose / gait / face / audio / player field anywhere in the schema** ? a
test
30     asserts this positively (deletion-at-source, §4c). Minors and biometrics are excluded.
```

```

31     - **CONTROL / pure-stdlib only.** The replay uses `experiment.CONTROL` (`is_learned() ==
False`)
32     ? no numpy / logistic-regression. Learned variants' near-threshold decisions flip
across CPUs,
33     so they can **never** anchor a cross-machine snapshot ($4d / D5).
34     - **Parse-compare, never byte-compare.** The gate `json.load`'s both sides: exact
equality on
35     int/structural fields, abs-tolerance `1e-6` on floats (CRLF/float-format safe).
36     - **Windowing is pinned per fixture.** The `(vt, mg, mwd, max_win)` params live in each
37     `provenance.json` and are passed explicitly to `build_candidates` ? the replay does
**not**
38     depend on the named `windowing.SERVED` constant, so a future SERVED retune can't
silently
39     churn fixtures.
40
41     ## Fixtures are GENERATED, never hand-edited
42
43     Do **not** hand-edit `expected.json` / `provenance.json` / the CSVs. They are produced
44     correct-by-construction by the CLI.
45
46     ```bash
47     # Generate the synthetic trajectory.csv + labels.csv for all built-in scenarios, then
48     # freeze expected.json + provenance.json + (re)write manifest.json:
49     python -m backend.eval.golden_fixtures --freeze-synthetic
50
51     # Replay all fixtures and parse-compare vs the committed expected.json (exit 0/1):
52     python -m backend.eval.golden_fixtures --check
53
54     # Re-freeze expected.json from the COMMITTED trajectory+labels (NO regeneration). ? This
can
55     # rubber-stamp a real regression ? review the visible diff and record a reason ($6):
56     python -m backend.eval.golden_fixtures --refresh-snapshot --all --reason "<why>"
57     python -m backend.eval.golden_fixtures --refresh-snapshot --slug clean_single_rally
--reason "<why>"
58     ```
59
60     ## Honest limits ($6 ? rides in the test pass/fail messages, not only here)
61
62     - A green suite proves only that the **deterministic post-trajectory transforms** are
unchanged
63     for the frozen cases. It does **not** cover the WASB/TrackNet detector, frame
decoding, the
64     AI/provider handoff, chunking, the ffmpeg stitch, DB persistence, or the API filter.
65     - Characterization is **behaviour-locking, not correctness** ? it freezes current
behaviour
66     *including current bugs*. "Passing" = "behaviour unchanged", never "behaviour
correct".
67     - **Synthetic-tier F1 measures plumbing correctness, NOT field quality.** Never cite a
synthetic
68     fixture's F1 in `QUALITY_ITERATIONS.md` or the paper as rally-detection performance.
69

```

21. user (2026-06-16T06:33:53.268Z)

```

45:| D4 | **Freeze seam = post-detector trajectory** *(design)* | workflow-1 synthesis | Widest
deterministic surface; reuse the shipped eval stack verbatim; the raw per-frame CSV is **never
committed** (only ~5 aggregate, fps/res-invariant shuttle scalars/window). |
54:**Core verdict (settled across IN/US/EU):** the derived numbers ? per-frame shuttle (x,y),
rally start/end intervals, audio/court/aggregate-person feature values ? **do NOT inherit the
source video's copyright.** They are uncopyrightable *facts/measurements*, not authored
expression, and not a §101 "derivative work."
96:- **Raw-trajectory exclusion** ? never commit the per-frame `(Frame,Visibility,X,Y)` CSV.

```

22. assistant (2026-06-16T06:34:20.098Z)

Now I have everything I need to render a verdict. Let me synthesize.

****The finding's factual headline is REFUTED, but it surfaces a genuine doc-internal contradiction.****

Refutation of the stated claim:

1. ****No `cleared\_real` fixture exists in the tree.**** All three committed fixtures (``clean_single_rally``, ``multi_rally_with_deadtime``, ``occlusion_break``) are ``kind="synthetic"``, ``license="synthetic"`` (manifest.json + provenance.json confirm). The finding's headline ? "Raw per-frame trajectory is committed verbatim ****for real fixtures****" ? is false: zero real fixtures are committed.
2. ****The cited evidence is synthetic, not real.**** ``clean_single_rally/trajectory.csv`` row ``1,1,723.164676,372.202099`` is from a ``kind="synthetic"`` fixture generated by deterministic formula (frame 0 is exactly the 1280x720 center ``640.000000,360.000000``, then a smooth parabolic arc). It is NOT a de Montjoye fingerprint of any real person ? there is no real source video behind it. The de Montjoye / re-ID argument the finding builds on does not apply to mathematically-generated coordinates.
3. ****`freeze\_real\_fixture` is tooling-only; it has not been run on real data.**** P1 (line 150) is "Tooling-only ... no real data committed"; P3 (the real-data commit, line 152) is status ? Todo. So the privacy-leak harm the finding describes (committing a real fingerprint) has not occurred and cannot occur until P3 + owner per-video clearance.

What the finding gets RIGHT (the partial-truth core):

- ``freeze_real_fixture`` does call ``write_trajectory_csv(rows, fdir/"trajectory.csv")`` (line 635) and ``_reset_trajectory_rows`` (lines 538-549) passes X/Y through untouched, shifting only the frame index. So IF a real fixture were ever frozen and committed via this tool, the full per-frame ``(Frame,Visibility,X,Y)`` would land in the tracked dir with the spatial fingerprint intact (only the timeline severed).
- That genuinely contradicts the doc's §4c line 96 (****Raw-trajectory exclusion**** ? never commit the per-frame ``(Frame,Visibility,X,Y)`` CSV"), §2 D4 line 45 ("the raw per-frame CSV is ****never committed****"), and the §4b/TM5 framing that Tier-0 is "5 shuttle scalars" only. The replay design fundamentally **REQUIRES** the committed ``trajectory.csv`` (it's the freeze-seam input ? ``replay_fixture`` reads ``fdir/"trajectory.csv"`` at line 311), so the "never commit the per-frame CSV" sentences are simply inconsistent with the shipped architecture. The fixtures README (line 14) openly lists ``trajectory.csv`` as a committed file, confirming the contradiction is within the doc, not the code.

So: the doc has a real, identifiable internal inconsistency (§4c/§2-D4/TM5 say the per-frame CSV is excluded; the actual M1+P1 design commits it). The proposed fix (b) ? reconcile the doc to state plainly that the full per-frame shuttle trajectory IS published, and re-do the §3/§5 non-attributability analysis on that basis ? is the correct remedy and is the substantive value of this finding.

Why isMustFix = false (this is a doc PR, gated, no real data):

- The PR under review is doc + tooling only; ****no real fixture is committed****, so there is no live privacy leak to "fix before relying on the suite." The suite runs entirely on synthetic data where re-ID is a non-issue.
- The refusal gate, time-origin reset, and ``_scan_forbidden`` are all present and working for the synthetic path. The doc-vs-code wording mismatch is a documentation-accuracy defect, not a suite-correctness or live-privacy defect.
- It becomes must-fix at the P3 gate (when a real fixture is actually frozen/committed) ? at that point the owner's 2026-06-16 risk acceptance was indeed made against doc text that says the raw CSV is excluded, so the doc must be corrected and re-accepted before any real-data commit. That is a future precondition, not a blocker for the current synthetic-only suite.

This is a "partial" verdict: the privacy-harm headline is refuted (no real data, synthetic-only, de Montjoye inapplicable), but the underlying doc-internal contradiction it points to (the doc says the per-frame CSV is never committed while the shipped design commits it) is real and worth fixing before P3.

23. user (2026-06-16T06:34:33.317Z)

Structured output provided successfully

24. assistant (2026-06-16T06:34:45.551Z)

I've completed the verification. Verdict: ****partial**** (isMustFix=false).

Summary of findings against the actual code in `C:\Users\avidu\Projects\badminton-highlight-indexer\claude\worktrees\gf-doc\`:

****The finding's privacy-leak headline is REFUTED:****

- No `cleared\_real` fixture exists. All three committed fixtures are `kind="synthetic"` / `license="synthetic"` ? confirmed in `eval\_baselines\fixtures\manifest.json` and `eval\_baselines\fixtures\clean\_single\_rally\provenance.json:6-9`.
- The cited "evidence" coordinate (`1,1,723.164676,372.202099`) is from a synthetic fixture: `eval\_baselines\fixtures\clean\_single\_rally\trajectory.csv:2` begins at the exact frame-center `640.000000,360.000000` and traces a deterministic parabola. There is no real person behind it, so the de Montjoye re-ID argument the finding builds on does not apply.
- `freeze\_real\_fixture` (`backend\eval\golden\_fixtures.py:557`) is tooling-only and has not been run on real data. P1 (doc line 150) is "no real data committed"; P3 (the actual real-data commit) is Todo, gated by the refusal gate + owner per-video clearance.

****The genuine core the finding correctly identifies (a doc-internal contradiction):****

- `freeze\_real\_fixture` does write the full per-frame CSV (`golden\_fixtures.py:635`), and `\_reset\_trajectory\_rows` (lines 538-549) passes X/Y through untouched. The replay design requires the committed `trajectory.csv` as its freeze-seam input (`replay\_fixture` reads it at `golden\_fixtures.py:311`), and the synthetic fixtures already commit it (fixtures README line 14).
- This contradicts the doc's own §4c (line 96), §2 D4 (line 45), and §4b/TM5 framing that the per-frame CSV is "never committed" / Tier-0 is "5 scalars" only. The doc is internally inconsistent with its shipped M1+P1 architecture.

****Why not must-fix:**** the PR is doc + tooling only with zero real data committed; the synthetic-only suite has no re-ID exposure, and the refusal gate / time-origin reset / `\_scan\_forbidden` all function. The doc-vs-code wording mismatch is a documentation-accuracy defect that becomes a hard precondition at the P3 gate (real-data commit) ? at which point proposed-fix (b) should be applied and the owner's risk acceptance re-confirmed against corrected text ? but it does not block relying on the current synthetic suite.